

Quantum History: An Automated Hypothesis-Engine Campaign

Bucket Foundation

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Abstract

We report one automated run of the Bucket Foundation history hypothesis engine [1] over the Quantum History corpus, with no human editing any hypothesis, evidence item, or score. From 16 sources and 161 evidence items, the engine generated 5990 distinct hypothesis addresses and carried 176 past its critic filter to scoring and the ranking tournament. A Chao1 coverage estimate over the generation seeds put the observed share of the address space at 0.023 to 0.031, with a Good-Turing missing-mass estimate of 0.040. 100.000% of survivors held a stable projected probability across the four prior-belief profiles this package ships. A k-fold holdout over 105 held-out ground-truth events (17 covered by this run’s own pre-holdout evidence) scored a Brier score of 0.423. The generator’s own target-blind check reported a non-consensus actor or mechanism proposal rate of 0.950 (steady against the previous run). We state every number in this report from the run’s own persisted artifacts, and list where those artifacts stop short of a full per-hypothesis accounting in 7.

1 Introduction

One run of the Bucket Foundation history hypothesis engine produced every result below, campaign `quantum-history`, run id `20260910T085020Z`, with no step in the run editing a hypothesis, evidence item, or score by hand. 4 names the engine-loop stages this run executed and their config; 5 and 6 state every quantitative outcome; 7 names what this run’s own artifacts do not carry.

2 Related work

The design and every definition this report depends on, the address space, the subjective-logic belief model, structural unknowns, and the engine loop itself, are stated and proved once, in the companion design paper [1]; this report adds no new definition of its own. The belief model’s own subjective-logic foundation is Josang’s [2].

3 Preliminaries

Every term below is defined in full in the design paper’s own Preliminaries and proved in its Lean listings; this section links each one to the module that implements it rather than restating the definition. An `opinion` (b, d, u, a) projects to a single `posterior` $P(h) = b + au$. A campaign’s `missing mass` and `coverage interval` come from a Chao1 estimate over its generation seeds. A `target-blind` check compares this run’s own non-consensus proposal rate against the run before it.

4 Method

This report’s own campaign follows the engine loop the design paper diagrams [1]: corpus ingestion, an optional extraction-ensemble pass, vocabulary growth from the unknown-unknown role, a combinatorial-plus-evidence-driven generation pass repeated across a fixed number of seeds, an LLM critic filter, a preservation critique against the shipped detectability table, subjective-logic belief scoring [2], a Swiss-style ranking tournament, robustness and surprise measurement across four prior-belief profiles, a Chao1 coverage estimate over the generation seeds, an optional discovery-date holdout calibration, and a self-report closing the run.

An extraction ensemble of three independent passes ran once over this corpus’s first document (`_CHAPTER`), producing 5 evidence items at an agreement score of 0.200 after escalating to the ensemble adjudicator.

This run used 3 generation seeds, 20 LLM-proposed placements per seed, a combinatorial sweep capped at 150 items per seed, a 400-hypothesis cap into scoring, 2 tournament rounds, and not recorded-level time resolution. 1 lists which model backed each engine-loop role for this run.

Table 1: Model backing each engine-loop role for this run.

Role	Model
critic	sonnet
extractor	haiku
generator	sonnet
judge	sonnet
meta_review	sonnet
preservation_critic	sonnet
referee	sonnet
self_report	sonnet
unknown_unknown	sonnet

5 Results

2 lists, for every time bin this run scored, how many placement hypotheses ranked into it and the highest-posterior survivor found there. 2 plots the full top- k posterior ranking per bin, and 1 plots the projected-posterior distribution across every distinct survivor these bins name. Beyond the per-bin view, this run’s survivors resolved into 40 competing-placement event groups (hypotheses sharing an object and place) and 0 competing-sequence pair groups.

The unknown-unknown role grew the vocabulary by 12 concepts this run (3 action, 4 actor, 2 mechanism, 2 object, 1 place). The Chao1 coverage estimate over this run’s generation seeds put the observed hypothesis count at 5990 against a Chao1 estimate of 226770.0, a missing-mass estimate of 0.040, and a 95% coverage band of [0.023, 0.031].

3 summarizes robustness across the four prior-belief profiles (`consensus`, `skeptic`, `fringe`, `uniform`); the underlying per-hypothesis projections are not themselves persisted by this run (7), so only the aggregate stable/unstable split is reported. The surprise rate, evidence naming no address any survivor materialized, was 0.130 (21 of 161 evidence items, by the same reasoning).

The evolver’s meta-review pass, reading this run’s whole survivor population and its opinions at once, reported the following. This is the role’s own output, quoted verbatim: paraphrasing a model’s structural read of its own run risks losing the detail a meta-review pass exists to surface.

The 176-hypothesis frontier looks diverse at the slot level (13+ actions, several mechanisms, ~15 p

Flags:

- Opinion collapse / lookup-table degeneracy: dozens of hypotheses that differ only in 'action' or
- Actor-locked authority prior: the 'a' parameter is drawn from a small fixed set of values that m
- Belief ceiling with one gamed exception: virtually the entire frontier is disbelief-dominated (b
- Flat uncertainty: u sits at ~0.009-0.012 for essentially all 176 hypotheses regardless of period
- Asymmetric exploration depth: Einstein-centric EPR/Solvay narratives receive exhaustive action-v
- Place slot under-exploration: the Planck/1900/blackbody-spectrum cluster almost never varies pla
- No high-confidence dissenting reading anywhere: for every canonical episode (Planck 1900, EPR 19
- Anachronism hypotheses (google-quantum-ai/sycamore, willow-below-threshold-result, shors-algorit

Recommended actions:

- Split the place='solvay-brussels'+interval=[1927,2009] mega-cluster into narrower sub-intervals
- Recombine fringe/low-authority actors (vendor-marketing, quantum-mysticism-popularizers, bell-ce
- Widen exploration of Bohr and Heisenberg-Schrodinger as primary actors for EPR-paradox and black
- Generalize the Planck-1900 and EPR-1935 clusters by replacing 'other-place'/'unknown-place' with
- Widen the next generation to explicitly target high-uncertainty (u) readings - construct hypothe
- Isolate and score the anachronism hypotheses (shors-algorithm, no-cloning-theorem, willow-below-
- Audit the opinion-generation function for the apparent (actor, place, mechanism, interval)-keyed

Table 2: Per-bin hypothesis counts and the top-ranked survivor in each.

Time bin	Ranked hypotheses	Top hypothesis	Top posterior
0	33	1c628f89fc80ce50	0.348
1	20	3330011c563a411f	0.061
2	46	02f5195f68a77fa2	0.549
3	67	f15b8d459226827f	0.162
4	2	ccf38da3cfbb7f5c	0.058
5	1	f73e818081406d8e	0.120
6	5	aab0a90b9a79b533	0.093
7	1	b147bc6bc7b74d0d	0.112
8	1	5fcef7c934a20309	0.144
9	0	n/a	not recorded
10	0	n/a	not recorded
11	0	n/a	not recorded
12	0	n/a	not recorded

Table 3: Robustness across the four prior-belief profiles (aggregate only, see 7).

Outcome	Count
Stable	176
Unstable	0
Total	176

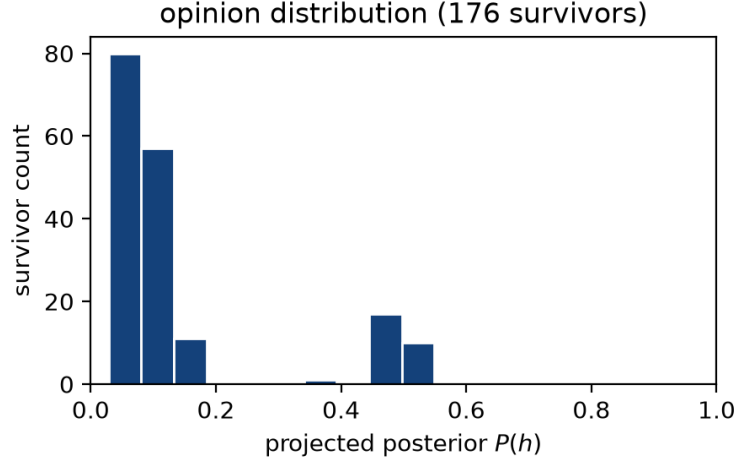


Figure 1: Projected-posterior distribution across every distinct survivor named in 2, generated by `figures/fig_opinion_histogram.py`.

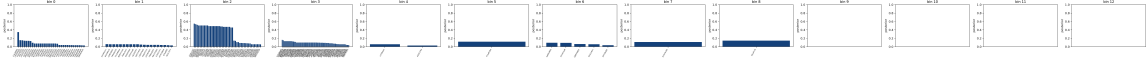


Figure 2: Top- k posterior ranking per time bin, generated by `figures/fig_bin_topk.py`.

6 Calibration

This run held out evidence via 5-fold cross-validation (seed 0), scoring 105 held-out ground-truth events (17 covered by this run’s own pre-holdout evidence) under belief constants $W = 2.0$, $\lambda = 0.50$. The resulting Brier score was 0.423. 4 lists every non-empty reliability bin; 3 plots the same bins against the diagonal a perfectly calibrated run would sit on.

Table 4: Discovery-date/ k -fold holdout reliability bins.

Predicted bin	Count	Mean predicted	Mean observed
[0.0, 0.1)	11	0.067	0.364
[0.1, 0.2)	7	0.132	0.857
[0.2, 0.3)	1	0.208	1.000
[0.3, 0.4)	2	0.311	1.000
[0.4, 0.5)	3	0.455	1.000
[0.5, 0.6)	3	0.518	0.333

7 Limitations

Two of this run’s own aggregates, robustness and surprise, are reported here only as the single scalar ‘MANIFEST.json’ carries (`robustness_stable_fraction`, `surprise_rate`); the per-hypothesis projections and the per-item surprise list that produced those scalars live only in the process’s own memory during a run and are not written to any file this paper can read. A future

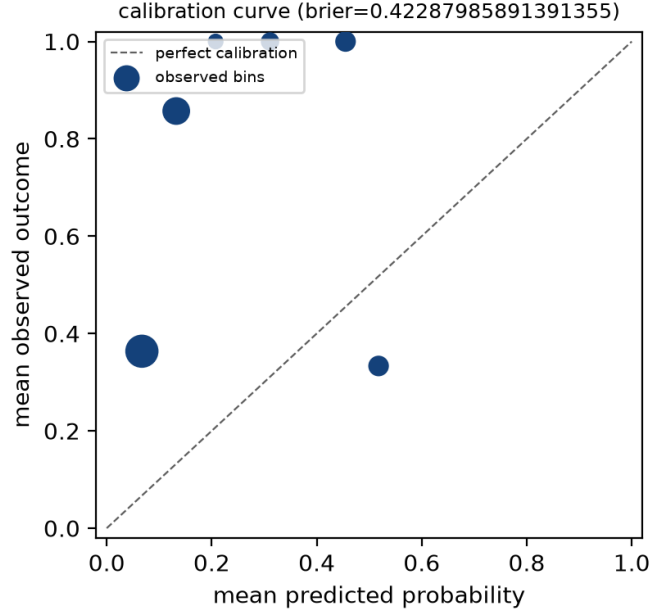


Figure 3: Discovery-date holdout reliability curve, generated by figures/fig_calibration_curve.py.

run that persists them would let a later paper over this same campaign carry both as real tables instead.

The run's own self-report follows, quoted verbatim, exactly as the self-report role returned it: its assumptions, which slot vocabularies it judged incomplete, its missing-mass estimate, its calibration summary, and its target-blind check.

```
{
  "assumptions": [
    "Evidence volume (161 evidence items, 16 sources) is treated as 'stable' relative to the prior",
    "The 11 vocab_added entries are treated as the full set of new terms introduced this run (no h",
    "coverage.missing_mass (0.03997) is treated as the primary point estimate; the reported covera",
    "robustness_stable_fraction=1.0 is read as a survivor-level stability metric (176/176 hypothes",
    "The meta_review's 'lookup-table degeneracy' finding is treated as directly relevant to vocabu",
  ],
  "incomplete_vocables": [
    "actor: all 4 new entries (Dowling and Milburn, Haroche and Wineland, Hanson group/TU Delft, E",
    "mechanism: only 2 additions (single-system/single-quantum control; event-ready entanglement s",
    "action: 3 additions (crystallized, awarded a prize, closed a loophole) are the clearest case",
    "object: 2 additions (second quantum revolution framing; loophole-free Bell test/NV-center swa",
    "place: only 1 addition (TU Delft) from a single ms pointer; meta_review separately flags that",
  ],
  "missing_mass_estimate": 0.04,
  "calibration_summary": "Calibration this run is weak: Brier score is 0.4229, well above the ~0.2",
  "target_blind_steady": true,
  "target_blind_note": "The generator's non-consensus actor/mechanism proposal rate is reported at",
}
```

References

- [1] Bucket Foundation. “A Hypothesis Engine over History”. Unpublished manuscript, Bucket Foundation. Held in this repository at `papers/history-hypothesis-engine/main.tex`; internal companion report, no DOI or arXiv id, see this file’s own header comment. 2026.
- [2] Audun Jøsang. *Subjective Logic: A Formalism for Reasoning Under Uncertainty*. Artificial Intelligence: Foundations, Theory, and Algorithms. Springer, 2016. DOI: [10.1007/978-3-319-42337-1](https://doi.org/10.1007/978-3-319-42337-1).

A Glossary

opinion <https://github.com/AGFarms/bucket-foundation/blob/master/tools/hypothesis-engine/hte/belief.py>
A subjective-logic opinion (b, d, u, a) : belief, disbelief, uncertainty mass, and base rate, fused from pooled supporting and refuting evidence weight against a total-ignorance constant W .

projected posterior <https://github.com/AGFarms/bucket-foundation/blob/master/tools/hypothesis-engine/hte/belief.py>
 $P(h) = b(h) + a(h) \cdot u(h)$, an opinion’s single-number projection, used to rank hypotheses within a time bin.

missing mass <https://github.com/AGFarms/bucket-foundation/blob/master/tools/hypothesis-engine/hte/unknowns.py>
The Good-Turing estimate of the address-space share made of hypotheses observed exactly once across a campaign’s generation seeds.

target-blind <https://github.com/AGFarms/bucket-foundation/blob/master/tools/hypothesis-engine/hte/runner.py>
A generation or judging pass that applies the identical standard regardless of whether the reading under test is the consensus or a fringe one; tracked run over run as the non-consensus actor/mechanism proposal rate.

coverage interval <https://github.com/AGFarms/bucket-foundation/blob/master/tools/hypothesis-engine/hte/unknowns.py>
A Chao1-based 95% band on how much of the address space this campaign’s generation seeds have observed, `hte.unknowns.coverage_interval`.